Invest in clean growth.
Stay selective on coal.

OUR THESIS: Clean expansion can coexist with fossil resilience.

Clean +25.4% vs all fossil +10.3% · 196 fixed countries · 2020–2024

CLEAN ENERGY

OVERWEIGHT

Diversified low-carbon generation

Participate in growing clean output; assess valuation and financing before security selection.

COAL

UNDERWEIGHT

U.S. domestic thermal focus

Consumption fell 14.4% through 2024; the 11.3% rebound in 2025 makes the call conditional.

3-YEAR HORIZON · NOV 2026–NOV 2029 | DASHBOARD: VERIFY → COMPARE → MONITOR

Energy Transition Analyst Team

Clean grew faster.
Fossil output still increased.

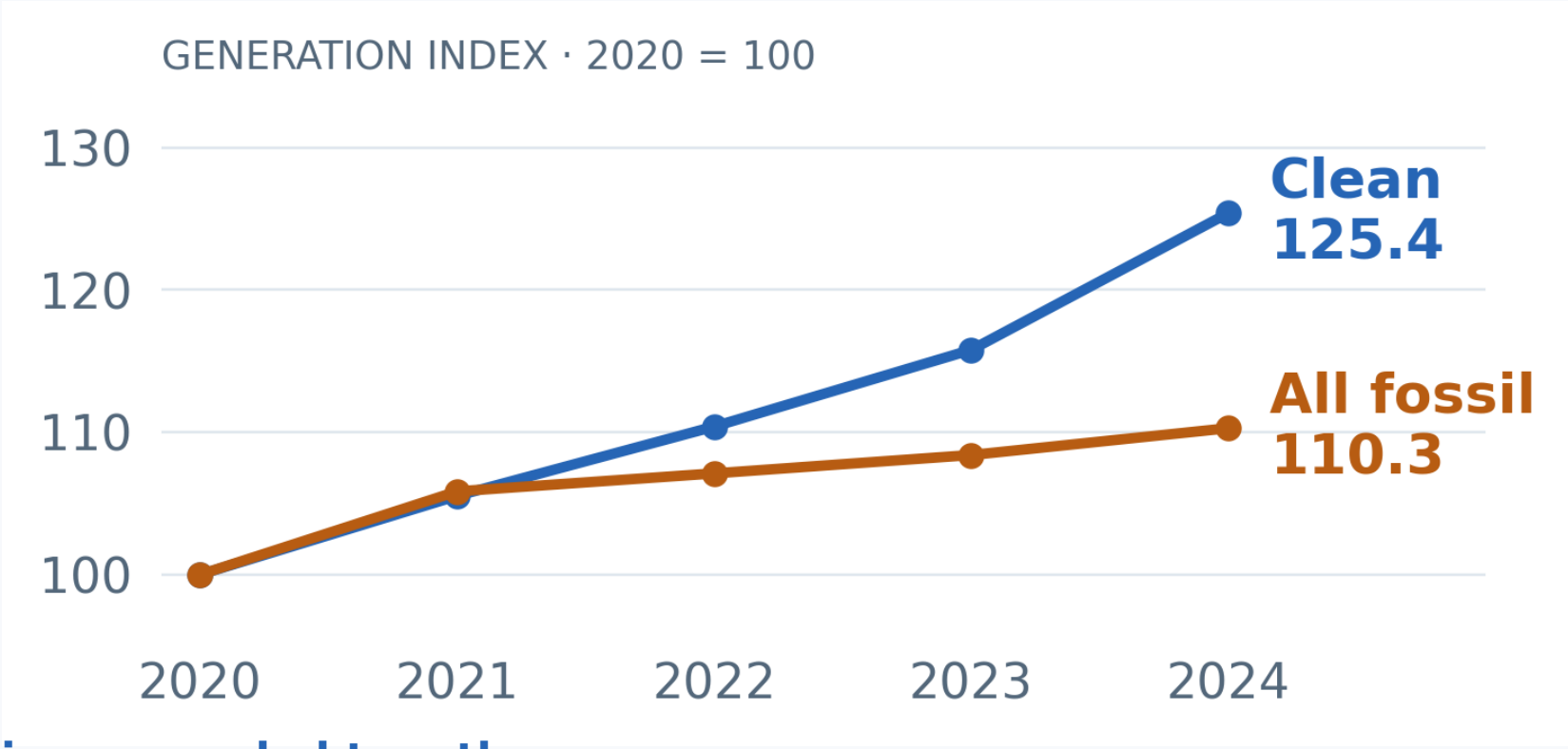
2020–2024 · the same 196 countries in every year

+25.4%

Clean generation

+10.3%

All-fossil generation



Clean and fossil generation expanded together.

Ember · fixed 196-country panel, 2020–2024 · E01–E02 · Clean = renewables + nuclear

The coal call needs a market context.

Three examples · clean and coal generation changes, 2020–2024

China

COEXISTENCE

Clean	**+50.9%**
Coal	**+18.4%**

Clean and coal expanded together.

United States

TRANSITION

Clean	**+14.1%**
Coal	**-15.6%**

Clean grew while coal generation fell.

Germany

DEMAND MATTERS

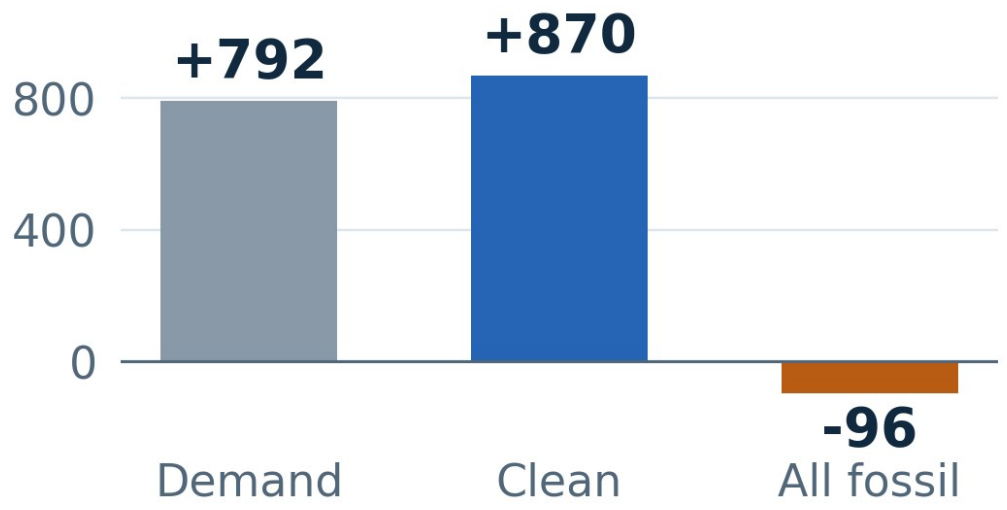
Clean	**-7.9%**
Coal	**-21.4%**

Demand also fell; coal decline alone does not establish substitution.

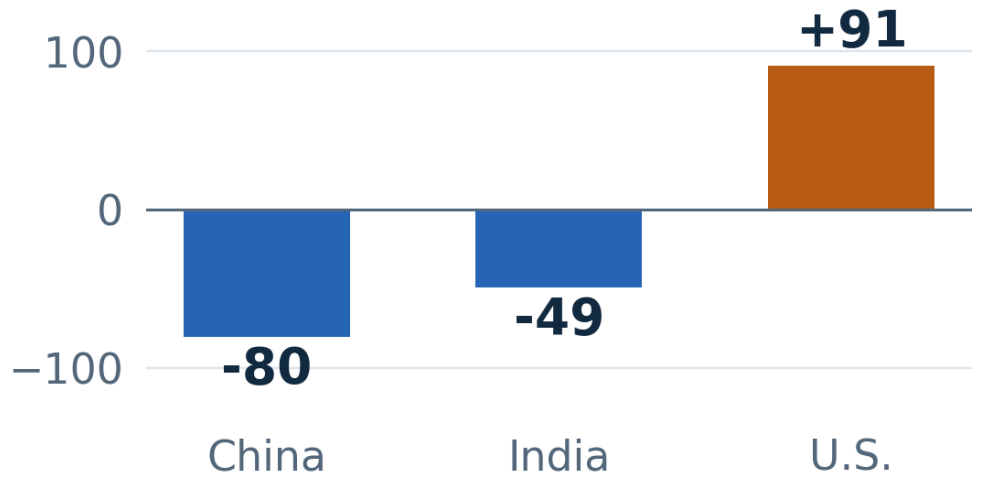
Similar changes in coal can have different investment implications.

2025 strengthened clean growth. Coal moved differently by market.

ELECTRICITY ADDITIONS · 91 COUNTRIES



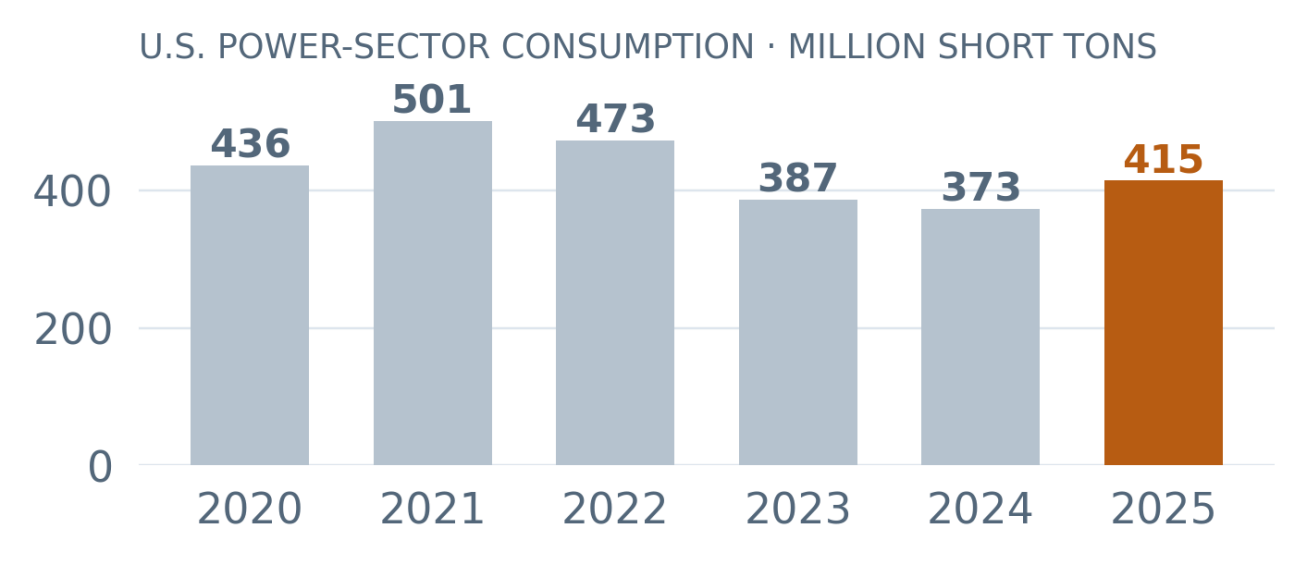
COAL GENERATION CHANGES · EI



Changes in TWh. The Ember sample represents 93.5% of observed 2024 extract generation, not the entire world.

The latest year supports clean growth and regional divergence.

U.S. coal recovered in 2025. Our underweight depends on persistence.



-14.4% 2020–2024 decline

+11.3% 2025 rebound

-4.7% 2025 vs 2020

Second measure: EI U.S. coal generation +91.2 TWh in 2025.

Same upward direction; generation and consumption have different units.

RECONSIDER if consumption stays above 2024 for two further annual releases.

Dashboard: United States → US electric power coal, 2020–2025

Historical decline supports caution; the rebound lowers conviction.

Match the country to the sector. Then test the investment economics.

UNITED STATES

Domestic thermal coal

CONDITIONAL UNDERWEIGHT

Power coal consumption −14.4% through 2024; +11.3% in 2025.

Check exposure to domestic power demand; monitor rebound persistence.

CHINA

Low-carbon generation

RESEARCH PRIORITY

Clean +50.9%; coal +18.4% through 2024: both expanded.

Diligence on generation owners: contracts, cash flow and financing.

GERMANY

Utilities & generation

RESEARCH PRIORITY

Clean −7.9%; coal −21.4%; demand −4.6% through 2024.

Check demand, electricity trade and revenue exposure before allocation.

China / Germany: diligence priorities, not buy ratings. All exposures need valuation and financial checks.

Favor clean growth; use country evidence to guide sector diligence.

What would change our view?

MONITOR	RECONSIDER WHEN…	RESPONSE
Coal recovery	U.S. consumption stays above 2024 for two further annual releases.	**Reduce the coal underweight.**
Clean momentum	Two comparable years without clean-share gains, alongside rising fossil output.	**Reassess the clean overweight.**
Investment economics	Weak project returns or financing pressure. Additional financial data is required.	**Review security selection.**

Data limits: changing coverage · source vintages · local EI download provenance

Use observable reversal conditions, and keep the data limits explicit.

Own clean growth.
Keep coal exposure selective.

OVERWEIGHT CLEAN ENERGY

Diversified low-carbon generation

UNDERWEIGHT COAL

U.S. domestic thermal focus

THREE-YEAR HORIZON | REVIEW THE POSITION AS THE EVIDENCE CHANGES

Three sources, three distinct measurement roles

CORE 1 · EMBER

1,138 country-year rows · 2020–2025
Demand, clean and all-fossil electricity in TWh. Clean includes renewables and nuclear.

CORE 2 · EIA

1,750 year/location/sector rows · 2020–2025
U.S. coal consumption in short tons. National sector 98 is kept separate from state base sectors.

OPTIONAL THIRD · ENERGY INSTITUTE

299,320 country/year/variable rows · 1965–2025
Adds coal-specific generation (TWh). Match country/year to Ember; compare U.S. direction with EIA, without equating tons to TWh.

Source: reports/data_quality_report.md; PROJECT_AUDIT.md; data/provenance/energy_institute.json

Preserve missingness. Prevent false comparisons.

PRESERVE SOURCE VALUES

Raw files are SHA-256 pinned. No source values overwritten; no duplicate keys found. Missing observations stay missing; explicit zeros remain zeros.

CONTROL COVERAGE & OVERLAP

196 fixed countries for 2020–2024; 91 for 2020–2025. Never sum overlapping EIA national, regional and state aggregates. EI world totals use WLD alone.

CHECK DEFINITIONS BEFORE CALCULATING

Validate keys, share ranges and accounting identities. EIA heat-content units are ambiguous: exclude those conversions from the thesis. Delivered price is a fuel cost, not a producer margin.

Source: reports/data_quality_report.md; tests/test_deliverables.py; analysis/verify_pitch.py

Faster clean growth survives a different baseline

2020 → 2024 · 196 FIXED COUNTRIES

Clean generation +25.4%
All-fossil generation +10.3%
Growth = (endpoint / baseline − 1) × 100.

2021 → 2024 · THE SAME 196 COUNTRIES

Clean generation +18.8%
All-fossil generation +4.2%
Direction persists after moving beyond the 2020 baseline.

WHAT THE CHECK DOES AND DOES NOT SHOW

The clean-growth lead is robust to this baseline choice. These are unequal-length cumulative windows, not annualized forecasts. The 91-country 2025 panel is a separate comparison.

Source: tables/baseline_sensitivity.csv · E01/E02/E10; tables/2025_watch_evidence.csv · E06

Classify pathways, then make the coal call specific

A TRANSPARENT COUNTRY SCREEN

Require clean and coal generation above 20 TWh at both endpoints. Apply directional changes beyond ±5% for 2020–2024. Sixteen material markets qualify; thresholds are analyst choices.

THE FINDING THAT CHANGES THE DECISION

China: clean +50.9%, coal +18.4%.
U.S.: clean +14.1%, coal −15.6%.
Germany: clean −7.9%, coal −21.4%; demand also fell.

TURN THE FINDING INTO AN ACTION

Favor diversified low-carbon generation exposure. Focus the coal underweight on U.S. domestic thermal producers. Review persistence of the U.S. rebound before extending the call.

Source: tables/material_country_pathways.csv; tables/country_review_all.csv · E07–E09

Measure growth, scale and the electricity mix

1 · GROWTH RATES & CAGR

2020–2024 CAGR: clean 5.8%; fossil 2.5%.
Why: annualize endpoint growth for comparison.
Pair rates with TWh to avoid small-base exaggeration.

2 · GENERATION-WEIGHTED SHARE

Clean share: 37.8% → 40.8%
Change: +3.06 percentage points.
Why: sum clean TWh / sum total TWh measures the mix of electricity produced.

3 · FIXED-PANEL COMPARISONS

Use the same countries and required metrics in every compared year.
Why: changing coverage can create false trends.
Keep the 196-country history and 91-country update separate.

Source: tables/method_examples.csv; tables/matched_2020_2024.csv; matched-panel membership tables

Check the explanation against the energy balance

4 · DEMAND-CHANGE ACCOUNTING

2024–2025 · 91 fixed countries
Demand +791.8 TWh
Clean +870.4; fossil -96.0; net imports +17.4 TWh.
Why: reconcile changes before interpreting them.

5 · BASELINE SENSITIVITY

Recompute growth from 2021 on the same 196-country panel. Clean still grows faster: +18.8% vs +4.2%.
Why: test whether the conclusion relies on pandemic-affected 2020.

6 · CROSS-SOURCE TRIANGULATION

Ember tests clean/fossil growth; EI isolates coal; EIA tests U.S. coal consumption.
Why: inspect the recommendation from distinct measures. Keep tons and TWh separate.

Source: tables/method_examples.csv; baseline_sensitivity.csv; 2025_watch_evidence.csv; SOURCES.md

Trace the recommendation back to the observations

1 · VERIFY THE HEADLINE

Data & evidence → Numbers supporting the presentation. Match E01/E02/E04/E05 to slides 2 and 5. Check source, units and fixed-panel scope before comparing.

2 · TEST THE INTERPRETATION

World → China, then Germany → expand demand/mix and coal charts. Compare 2020 and 2024. Research → Coal shows EI's world total separately from matched samples.

3 · CHALLENGE AND REPRODUCE THE CALL

United States → national consumption and delivered price. Show the 2025 rebound. Return to Data & evidence for coverage, claim references and downloadable observations.

Blue fill means a higher selected value.



Low → high selected value · read the filter and units first

FILTER	WHAT THE COLOR / SCOPE MEANS	WHAT TO INSPECT NEXT
Clean electricity share · %	Darker blue = higher clean share	Open clean/fossil output and demand; check growth and economics.
Renewable generation · TWh	Darker blue = more renewable output	Check output growth and scale, then project valuation.
Electricity demand · TWh	Darker blue = more electricity demand	Compare demand additions with clean and fossil additions.
Carbon intensity · gCO2e/kWh	Darker blue = higher emissions per kWh	Inspect mix and emissions. Darker here = more carbon-intensive.

Dark blue + red outline can mean a high clean share AND rising coal generation. Inspect both histories.

Missing fill means unreported, not zero. Compare tooltip values: scales change with year and filter.

Source: dashboard/src/data/types.ts; components/Legend.tsx; components/USMap.tsx; viz/layerValue.ts

Orange fill shows coal scale or dependence.



Low → high selected value · read the filter and units first

FILTER	WHAT THE COLOR / SCOPE MEANS	WHAT TO INSPECT NEXT
Coal-fired generation · TWh	Darker orange = more coal electricity	Open coal and demand histories; distinguish size from growth.
Coal share of generation · %	Darker orange = greater coal dependence	Compare coal TWh and clean share; dependence differs from scale.
Coal production · Mt	Darker orange = more coal produced	Check producer exposure; exports require trade evidence.
Coal energy supply (TES) · EJ	Darker orange = more coal energy supply	Inspect energy-system exposure, including non-power uses.

Red outline = coal generation rose vs prior year; blue = fell. Direction prompts research, not a long/short trade.

Missing fill means unreported, not zero. Compare tooltip values: scales change with year and filter.

Source: dashboard/src/data/types.ts; components/Legend.tsx; components/USMap.tsx; viz/layerValue.ts

U.S. coal & clean: national first, then states.



State map: low → high power coal consumption

FILTER	WHAT THE COLOR / SCOPE MEANS	WHAT TO INSPECT NEXT
Generation · TWh	Blue = clean; red = coal	Compare national output histories from Ember and EI.
Share · %	Each source’s reported generation share	Distinguish electricity mix from absolute output.
Priority coal-demand markets	EIA state power consumption, sectors 1+2+3	Rank market size and annual growth; open a state history.
Optional state map & Map key	Darker orange = more coal consumed	Inspect coal customers. No state clean data is inferred.

2025: Texas, Missouri and Indiana lead reported power coal demand and all grew. Prioritize supplier exposure research.

Missing fill means unreported, not zero. Compare tooltip values: scales change with year and filter.

Source: dashboard/src/data/types.ts; components/Legend.tsx; components/USMap.tsx; viz/layerValue.ts